

Master of Science (IT) SEM – 10



Lok Jagruti Kendra University
University with a Difference

LJ Institute of Computer Applications

RetailIQ - Smart Retail Operations & Analytics Platform

Guided By:

Developed By:

<i>Sr.</i>	<i>Enrollment</i>	<i>Name</i>
<i>1</i>	<i>21004500210065</i>	<i>Arnold Macwan</i>
<i>2</i>	<i>21004500210094</i>	<i>Manthan Parmar</i>

Submitted To:

LJ Institute of Computer Applications

LJ Campus, Nr. Sanand Cross Road, S.G Road,

Ahmedabad – 382110

Phone: +91 9099063417



Lok Jagruti Kendra University
University with a Difference

Certificate

Enrollment: 21004500210065

This is to certify that **Arnold Macwan** of MSc. IT, Semester 10, Roll no. B28 has satisfactorily completed his project with the title of **“RetailIQ - Smart Retail Operations & Analytics Platform”** under the supervision of guide.

Internal Guide:

HOD:



Lok Jagruti Kendra University
University with a Difference

Certificate

Enrollment: 21004500210094

This is to certify that **Manthan Parmar** of MSc. IT, Semester 10, Roll no. B41 has satisfactorily completed his project with the title of **“RetailIQ - Smart Retail Operations & Analytics Platform”** under the supervision of guide.

Internal Guide:

HOD:

Plagiarism Declaration

To whom so ever it may concern I/We, confirm that Project (document/PPT or Code) is my own work, is not copied from any other person's work (published or unpublished or generated using CHATGPT/AI, and has not previously submitted for assessment either at University or elsewhere. We confirm that we have read and understood the rules regulations on plagiarism in LJ University.

<i>Sr.</i>	<i>Enrollment</i>	<i>Name</i>
<i>1</i>	<i>21004500210065</i>	<i>Arnold Macwan</i>
<i>2</i>	<i>21004500210094</i>	<i>Manthan Parmar</i>



Lok Jagruti Kendra University
University with a Difference

Commitment Form

I/we assure that following are the components on which we worked in the said project titled **RetailIQ - Smart Retail Operations & Analytics Platform**. Further we confirm that, we have read and understood the rules and regulations of UFM in LJU.

Student 1.

Enrollment: 21004500210065

Div.: A

Roll No.: B28

Name: Arnold Macwan

Sr.	Component	Sr.	Component
1.	Role based access control	6.	Payment Interface
2.	Secured Routing & Session management	7.	Ask Supplies Orders
3.	Analytical Dashboard	8.	Bid Orders (supplier)
4.	Manage stores	9.	Invoice module
5.	Inventory management	10.	Stock-out Alert



Lok Jagruti Kendra University
University with a Difference

Commitment Form

I/we assure that following are the components on which we worked in the said project titled **RetailIQ - Smart Retail Operations & Analytics Platform**. Further we confirm that, we have read and understood the rules and regulations of UFM in LJU.

Student 2.

Enrollment: 21004500210094 Div.: A Roll No.: B41

Name: Manthan Parmar

Sr.	Component	Sr.	Component
1.	Login - Signup	6.	Ask orders Analytics
2.	Feedbacks	7.	Bid Payment Records
3.	Profile management	8.	Product Catalogs
4.	Supplier Dashboard	9.	Order Billings
5.	Product Category management		



Lok Jagruti Kendra University
University with a Difference

Acknowledgement

I would like to take this opportunity to express my sincere gratitude and appreciation to all those who have supported me in the successful completion of my Semester 10 project. First and foremost, I would like to extend my heartfelt thanks to Dr. Jignesh Doshi, Head of the Department, M.Sc. (Information Technology), LJ School of Computer Applications, LJ University, for his continuous encouragement, expert guidance, and invaluable support throughout the duration of this project.

I am extremely thankful to my internal guide, Mr./Ms. Virendra Ingle, for providing insightful feedback, motivation, and guidance at every stage of the project. I would also like to acknowledge the contribution of my external guide, whose expertise and suggestions greatly enhanced the quality of my work.

I sincerely appreciate the efforts of all faculty members and lab instructors who provided assistance and support whenever needed. Lastly, I am grateful to my family and friends for their constant support, motivation, and encouragement throughout.

Semester 10.

Project title: RetailIQ - Smart Retail Operations & Analytics Platform

<i>Sr.</i>	<i>Enrollment</i>	<i>Name</i>
<i>1</i>	<i>21004500210065</i>	<i>Arnold Macwan</i>
<i>2</i>	<i>21004500210094</i>	<i>Manthan Parmar</i>

Sr.	Table of Content	Page No.
1	Introduction	09
	1.1 Existing system	09
	1.2 Need for new system	10
	1.3 Objective	10
	1.4 Problem Definition	11
	1.5 Scope of project	11-12
	1.6 Project Profile	12
	1.7 Core components	13
	1.8 Assumptions & constraints	13-14
	1.9 Advantages & Limitations	14
2	Requirement determination & analysis	15
	2.1 Requirement determination	15-16
	2.2 Targeted users	17
3	System design	18
	3.1 Use case diagram	18
	3.2 Class diagram	19
	3.3 Activity diagram	20
	3.4 Sequence diagram	21
	3.5 Data dictionary	22
4	Development	
	4.1 coding standard	
	4.2 Screenshots	
	4.3 Implementation of business logic	
	4.4 Navigation flow	
	4.5 Server-side validation (optional if needed)	
	4.6 Client-side validation (optional if needed)	
5	Agile documentations	
	5.1 Project charter	
	5.2 Roadmap/Schedule	
	5.3 User story	
	5.4 Release plan	
	5.5 Sprint backlog	
	5.6 Test plan	
6	Proposed enhancement	
7	Conclusion	

1. Introduction

The retail sector is undergoing in a massive **digital transformation** driven by the increasing need for efficiency, intelligent decision making and **data-driven operations**. Meanwhile the large organization leverage sophisticated use of AI-powered tools and automation techniques, but the small-medium size enterprises often rely manual processes and operations and traditional **POS – point of sale** systems.

RetailIQ is designed to bridge this technological gap by providing a smart and unified platform which combines all the management and processes with advanced analytics. The system empowers local shops, supermarkets, and fashion stores to manage inventory, track sales, monitor supplier performance, and understand purchase patterns — all in a centralized ecosystem.

1.1. Existing System

In the current retail environment, most small and medium-sized stores rely on traditional **Point-of-Sale** (POS) software or manual methods to manage their operations. These existing systems primarily focus on **basic functionalities** such as billing, inventory updates, and transaction recording. While effective for routine tasks, they **lack intelligent decision-support** capabilities. Retailers must manually analyse sales data, estimate future demand, and track stock levels, which often leads to *inaccuracies* and *inefficiencies*.

Overall, the existing system is **transactional** rather than analytical. It records data but does not help retailers interpret it for **strategic decisions**. As a result, retail operations become reactive, inefficient, and less competitive in the modern digital market.

1.2. Need for new system

With increasing competition and rapidly changing consumer behavior, retailers require more than just POS software—they need an **intelligent platform** that can transform raw data into actionable insights. This necessity creates a strong demand for a system like RetailIQ, which integrates **conventional** retail management with advanced analytics and AI-driven predictions.

The system is needed to:

- Enhance decision making
- Prevent stock-out
- Understand customer behavior
- Streamline business management
- Reduce human error

The **need for the system** arises from the growing demand for smart, analytical retail operations that can support better decision-making, reduce costs, and increase competitiveness.

1.3. Objective

The main objective of RetailIQ is to develop an intelligent, integrated platform that enhances retail operations through advanced data analytics and AI-powered decision support. The specific objectives include:

- Automate & optimize inventory management
- Advanced sales & profit analysis
- Data-driven decision making
- Enhance customer relations
- Improve accuracy
- Scalable and unified platform

Build an AI-driven, analytics-focused system that enhances decisions, optimizes inventory, improves sales strategies, and empowers retailers with insights previously accessible only to large enterprises.

1.4. Problem statement

Retail businesses, especially small and mid-sized stores, generate large volumes of transactional data through daily sales, purchases, and inventory movements. However, most retailers lack the tools and expertise to analyze this data effectively. These limitations prevent retailers from operating efficiently, understanding their customers, or forecasting future business needs. Therefore, the problem is the absence of a unified, smart retail management system that not only handles operations but also provides predictive analytics, actionable insights, and intelligent decision support for modern retail environments.

1.5. Scope of the project

The scope of RetailIQ encompasses the design, development, implementation, and evaluation of an intelligent retail management and analytics platform aimed at empowering small and mid-sized retail businesses. The project covers functional, non-functional, analytical, and technological aspects as described below.

- I. Operational scope
 - a. Inventory management
 - b. Sales & Billing integration
 - c. Supplier & purchase management
- II. Analytical scope
 - a. Demand forecasting
 - b. Market basket analysis
 - c. Stock prediction
 - d. Interactive dashboards
- III. User Scope
 - The system is intended for:
 - a. Local shop owners
 - b. Vendors, mini-markets

- c. Retails stores

IV. Technical scope

- a. Frontend interface
- b. Backend services
- c. API services
- d. Database systems
- e. Machine learning pipelines

V. Out of scope

- a. Multi-store management
- b. Real-time or barcode hardware systems
- c. Integration with third part e-commerce systems

The project will deliver a functional, scalable platform capable of transforming traditional retail data into meaningful business intelligence.

1.6. Project profile

Project Name	RetailIQ
Frontend	HTML, CSS, JavaScript, ReactJS
Backend	ExpressJS, NodeJS
Database	MongoDB, MySQL
ML Services	Python, FastAPI
Tools	VS Code, MS-Office

1.7. Core Components

Admin	Supplier	Customer	Store manager
• Login • Manage suppliers • Manage stores • Manage permissions • Analytics & monitoring • Pricing & Discounts • Reports • Logout	• Login • Manage purchases • Products & pricing • Communication • Profile management • Order analytics • Logout	• Login • Registration • Profile management • Past purchases • Feedbacks • Orders & Wishlist • Logout	• Login • Profile Management • Inventory management • Sales & Billing • Manage supplies • Sales dashboard • Store level Analytics • Logout

1.8. Assumptions & Constraints

➤ Assumptions

- Uploaded datasets will not contain malicious or corrupted files that break the system.
- Users will provide timely feedback during each iteration for improvements.
- The system will operate in an environment with stable internet connectivity, as it relies on external services/APIs.
- Users will have basic digital literacy to operate the dashboard, upload files, and understand analytical insights.
- Retailers will provide accurate sales, inventory, and purchase data, either through POS exports or manual uploads.
- Other historic data will be used for training model for stock/sales analysis and prediction if retailer does not have sufficient data.

➤ **Constraints**

- Computational resources (RAM, CPU) may restrict training of large ML models.
- Retailers unfamiliar with technology may require training to fully utilize analytics features.
- Lack of sufficient historical data may reduce forecasting performance.
- System performance may depend on server capabilities (processing time for large datasets).
- Web browser compatibility may limit advanced visualizations.
- The ML model accuracy may vary based on dataset size, seasonal variations, and store type.

1.9. Advantages & Limitations

➤ **Advantages**

- Data-driven decision making
- Improved inventory management
- Enhanced profitability
- Better customer segmentation and understanding
- Enhanced stock management

➤ **Limitation**

- Accuracy depends on data quality
- Can't capture external market factors for model training
- Limited real-time analytics
- Lack of back-up service
- Multilingual support
- Low maintenance

2. Requirement determination & Analysis

Requirement determination and analysis is the process of gathering, studying, and validating the needs of users, stakeholders, and the system itself. This phase ensures that the project's objectives align with real-world retail operations and that the final system is both technically feasible and user-oriented.

For RetailIQ, requirement analysis focuses on understanding how retailers currently manage operations, what limitations they face, and what features are essential to build an effective, intelligent retail analytics platform.

2.1. Requirement determination

Requirement determination and gathering has been done through several methods and techniques such as:

- Interviews
- Observations of existing system
- Case studies
- Surveys/Feedbacks
- Industry research

2.2. Requirement analysis

After gathering information from different sources, the requirements are analyzed and broken into categories:

I. Functional requirement

a. Inventory management

- Add, Read, Update or Delete products
- Track stock levels automatically
- Identifying slow moving stocks
- Low-High stock alerts

b. Sales management

- Upload and handle sales data
- Generate reports
- Identify sales trends

- Billing and sales integration
- c. Supplier management
 - Register suppliers
 - Manage purchase history
 - Track supplies
- d. AI-ML module
 - Demand forecast
 - Market basket analysis
 - Stock/sales/profit prediction
- e. Dashboard & Reporting
 - Interactive visuals & charts
 - KPI insights
- f. User management
 - Login, Register authentication
 - Role based access control
 - Purchases
 - Interests

II. Non-functional requirement

- a. Performance requirement
- b. Reliability
- c. Usability
- d. Security
- e. Scalability

III. User requirement

- a. Easy product/inventory management
- b. Sales analytics
- c. Recommendation
- d. Reports
- e. Accessible

IV. System requirement

- a. Standard PC with windows 10+ and stable internet connection
- b. Python installed if needed in local PC
- c. Cloude storage or other storage devices to store sales data

2.3. Targeted Users

The system primarily targets users who manage retail activities but lack access to advanced analytics or enterprise-level retail solutions.

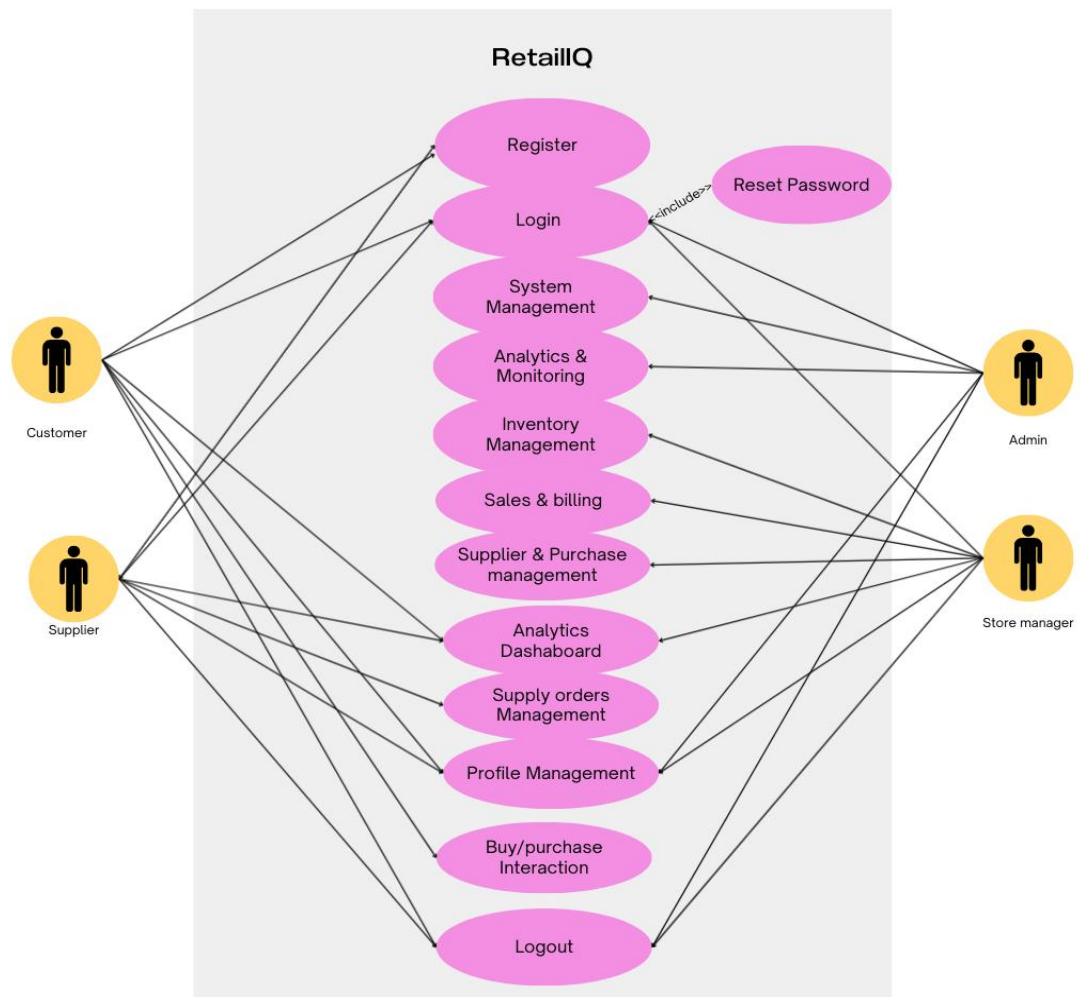
- Retail shop owners
- Supermarket store managers
- Retail data analyst
- Store managers & operational staff
- Suppliers & vendors (passive users)

With context of login or role-based users the system allows following users:

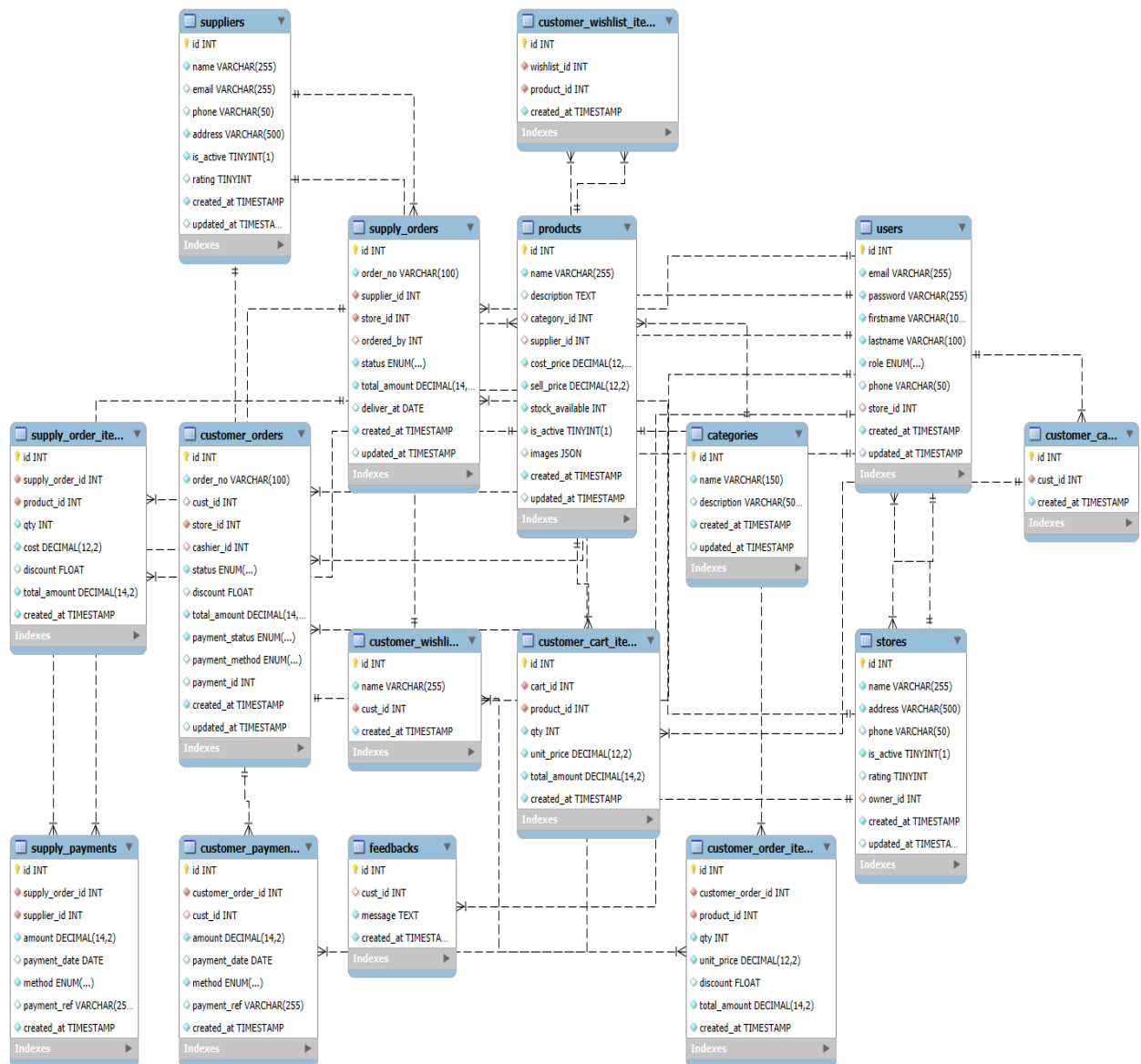
- Admin
- Customer
- Store Manager/staff
- Supplier

3. System Design

3.1. Use Case Diagram

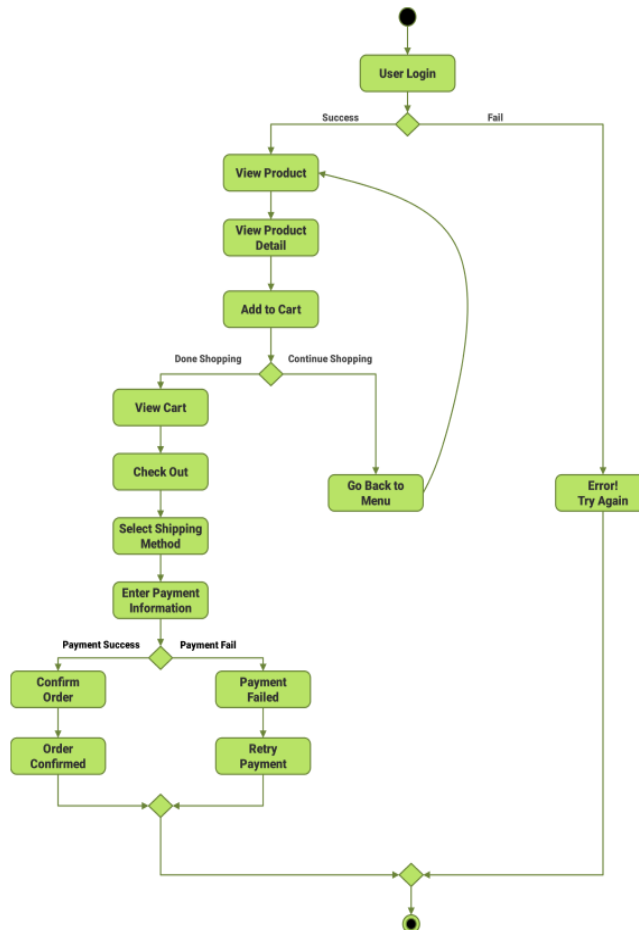


3.2. Class Diagram



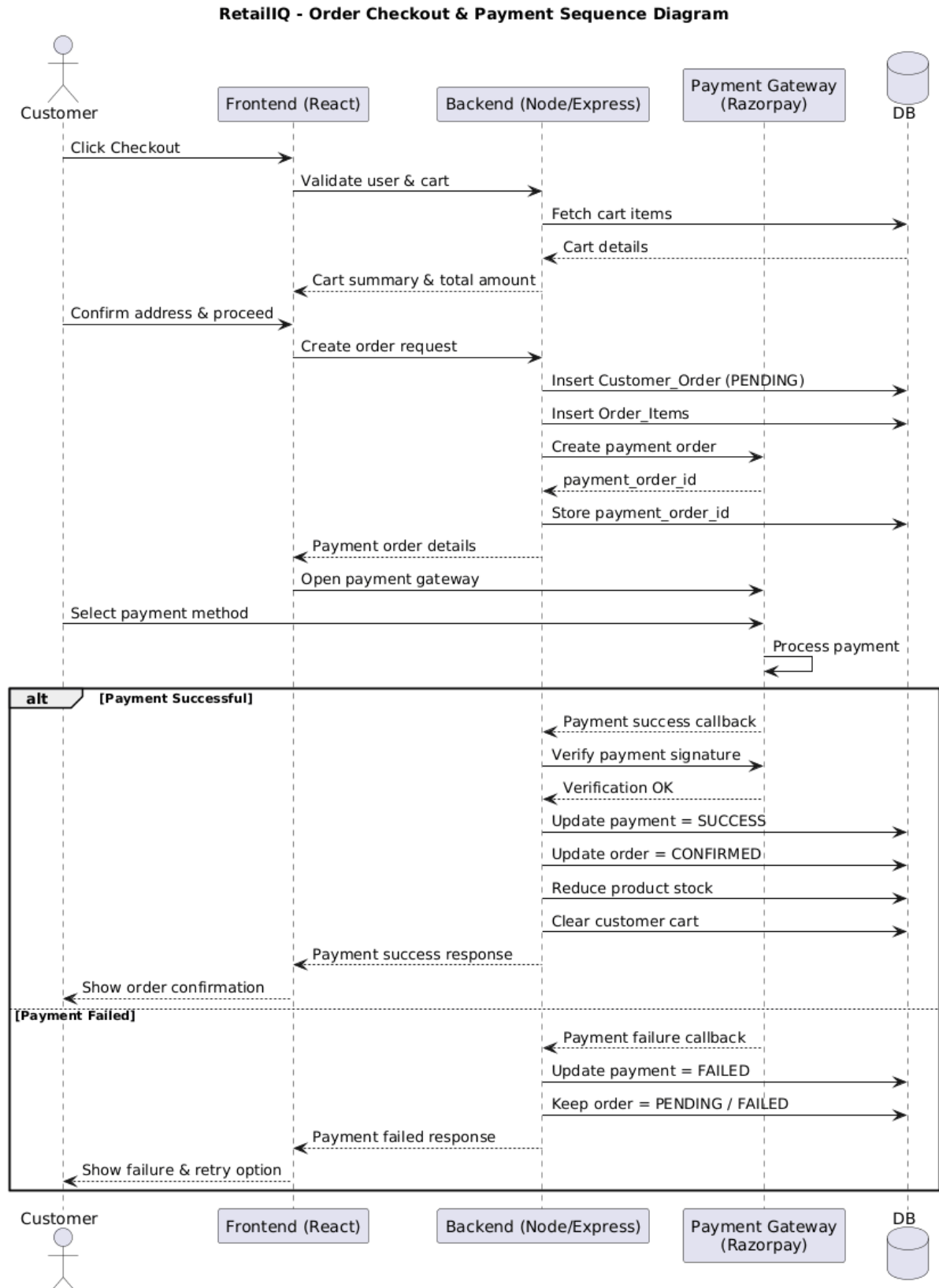
3.3. Activity Diagram

- Order checkout and payment activity



3.4. Sequence Diagram

- Order checkout and payment activity



3.5. Data Dictionary

1. Users

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Email	Varchar	Unique	abc@gmail.com	Email id of the user
Password	Varchar	Not null	123456	Hash password
Firstname	Varchar	Not null	John	Firstname of the user
Lastname	Varchar	Not null	Doe	Lastname of the user
Role	Enum (admin, customer, store manager)	Not null	Admin	Role of the user for access
Phone	Varchar	Unique	82**7**2*8	Phone number
Store_id	Int	Foreign key (refers store table)	001	Store id which belongs to a store manager
CreatedAt	Timestamp	Default	20-02-2025	Timestamp of record creation

2. Categories

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Name	Varchar	Unique	Men	Category for the products
Description	Varchar	-	All clothes for men	Summary of the category
CreatedAt	Timestamp	Default	21-12-2025	Date of creation

3. Products

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Name	Varchar	Unique	JoggersX	Name of the product
Description	Varchar	-	Skinny fit jogger	Summary of the product
CreatedAt	Timestamp	Default	21-12-2025	Date of creation
CategoryId	Int	Foreign key (refers category table)	011	A Category belongs to the product
SupplierId	Int	Foreign key (refers supplier table)	121	A supplier belongs to product

CostPrice	Decimal	Not null	\$ 499	Price of cost from user
SellPrice	Decimal	Not null	\$ 599	Price of sale
Stock_available	Int	Not null	50	Quantity of stock available
isActive	Boolean	Not null	True	Status of product
Images	Array	-	abc.jpeg	Images for the products

4. Suppliers

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Name	Varchar	Unique	Torrent supplies Ltd.	Name of the supplier
Email	Varchar	Unique	ts@gmail.com	Email of the supplier
CreatedAt	Timestamp	Default	21-12-2025	Date of creation
Phone	Varchar	Unique	7820**41*6	Phone/contact number
Address	Varchar	Not null	Avenue complex, NYC, USA	Address of the supplier
isActive	Boolean	Not null	False	Status of the supplier
Rating	Enum (1-5)	-	4	Rating for supplier

5. Stores

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Name	Varchar	Unique	Fista	Name of the store
Address	Varchar	Not null	Siteni complex, NYC, USA	Address of the store
CreatedAt	Timestamp	Default	21-12-2025	Date of creation
Phone	Varchar	Unique	7820**41*6	Phone/contact number

isActive	Boolean	Not null	False	Status of the supplier
Rating	Enum (1-5)	-	4	Rating for supplier
ownerId	Int	Foreign key (refers user table)	0131	A user/owner belongs to store

6. Supply_Orders

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
OrderNo	String	Unique	PO-52045	Unique identifier for order
supplierId	Int	Foreign key (refers suppliers table)	125	Supplier belongs to supply order
storeId	Int	Foreign key (refers store table)	111	Store from order belongs
orderedBy	Int	Foreign key (refers user table)	101	User belongs to order
Status	Enum	Not null	Sent	Status of the order
TotalAmount	Decimal	Not null	\$ 4500	Total amount for the order
deliverAt	Date	Not null	12-05-26	Date of delivery
createdAt	Datetime	Default	11-04-26	Record creation

7. Supply_Orders_items

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Supply_order Id	Int	Foreign key (refers supply_orders)	500	Supply Items belongs to supply
productid	Int	Foreign key (refers products)	511	Product belongs order
Qty	Int	Not null	50	Quantity of product
Cost	decimal	Not null	\$ 50	Price per unit of the product
Discount	Float	-	5.5	Discount if any
Total_amount	Decimal	Not null	\$ 2500	Total amount for order
createdAt	Datetime	Default	11-04-26	Record creation

8. Supply_Payment

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Supply_order Id	Int	Foreign key (refers supply_orders)	500	Supply Items belongs to supply
supplierId	Int	Foreign key (refers supply_orders)	111	Supplier belongs to order
Amount	Decimal	Not null	\$ 32500	Amount of payment
Payment_date	Date	Default	12-05-26	Date of payment
Method	Enum	Not null	IMPS	Transaction method
paymentRef	String	Not null	PayId: 56512	Reference of payment
createdAt	Datetime	Default	11-04-26	Record creation

9. Customer_Orders

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
orderNo	String	Unique	CO-52146	Unique no. of order
custId	Int	Foreign key (refers user table)	111	User belongs to order
storeId	Int	Foreign key (refers store table)	233	Store belongs order
Cashier	Int	Foreign key (refers user table)	166	Order created by
Status	Enum	Not null	Pending	Status of the order
Discount	Float	-	5.56	Discount if any
createdAt	Datetime	Default	11-04-26	Record creation
totalAmout	Decimal	Not null	\$ 4500	Total amount of the order
Payment_status	Enum	Not null	Pending	Status of the payment
paymentMethod	Enum	Not null	Pending	Method of payment
paymentId	Int	Foreign key (refers customer Payment)	52242	Payment reference

10.Customer_Orders_items

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Cust_orderId	Int	Foreign key (refers customer_orders)	500	order Items belongs to customer
productid	Int	Foreign key (refers products)	511	Product belongs order
Qty	Int	Not null	50	Quantity of product
Cost	decimal	Not null	\$ 50	Price per unit of the product
Discount	Float	-	5.5	Discount if any
Total_amount	Decimal	Not null	\$ 2500	Total amount for order
createdAt	Datetime	Default	11-04-26	Record creation

11.Customer_Payment

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
customer_orderId	Int	Foreign key (refers customer_orders)	500	order Items belongs to customer order
custId	Int	Foreign key (refers user)	111	customer belongs to order
Amount	Decimal	Not null	\$ 32500	Amount of payment
Payment_date	Date	Default	12-05-26	Date of payment
Method	Enum	Not null	IMPS	Transaction method
paymentRef	String	Not null	PayId: 56512	Reference of payment
createdAt	Datetime	Default	11-04-26	Record creation

12.Customer_wishlist

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
Name	String	Not null	My wishlist	Wishlist name
custId	Int	Foreign key (refers user)	111	customer belongs to order
createdAt	Datetime	Default	11-04-26	Record creation

13.Customer_wishlist_items

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
wishlistId	Int	Foreign key (refers wishlist)	699	Wishlist reference
productId	Int	Foreign key (refers product)	155	Product belongs wishlist
createdAt	Datetime	Default	11-04-26	Record creation

14.Customer_cart

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
custId	Int	Foreign key (refers users)	121	User belongs cart
createdAt	Datetime	Default	11-04-26	Record creation

15.Customer_cart_items

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
cartId	Int	Foreign key (refers customer cart)	566	Cart items belong to cart
productId	Int	Foreign key (refers product)	155	Product belongs cart
createdAt	Datetime	Default	11-04-26	Record creation
Qty	Int	Not null	5	Quantity of product
Unit_price	Decimal	Not null	\$ 25	Price per product
Total_amount	Decimal	Not null	\$ 125	Total amount to be paid

16.Feedbacks

Name	Type	Constraints	Example	description
Id	Int	Primary key	001	Table identifier
custId	Int	Foreign key (refers user)	121	User belongs feedback
Message	Text	Not null	“Good”	Message for feedback
createdAt	Datetime	Default	11-04-26	Record creation

4. Development

4.1. Coding standard

- *Project structure*
Use clear and descriptive names for the whole directory of the project and also recommend the naming conventions rules for the variables and functions names.
- *Code version management*
Use a version control system (i.e., GitHub) for the project to collaborate effectively.
- *UI Design*
Design a user-friendly interface so that user can use the system and understand it easily
- *Error Handling*
Implement an error free code and use error handling mechanism for user interactions
- *Further development*
Maintain the system as well conduct regular reviews that genuine feedback for the system.
Always look for the further improvement of the system that can make the system work more accurate and reliable

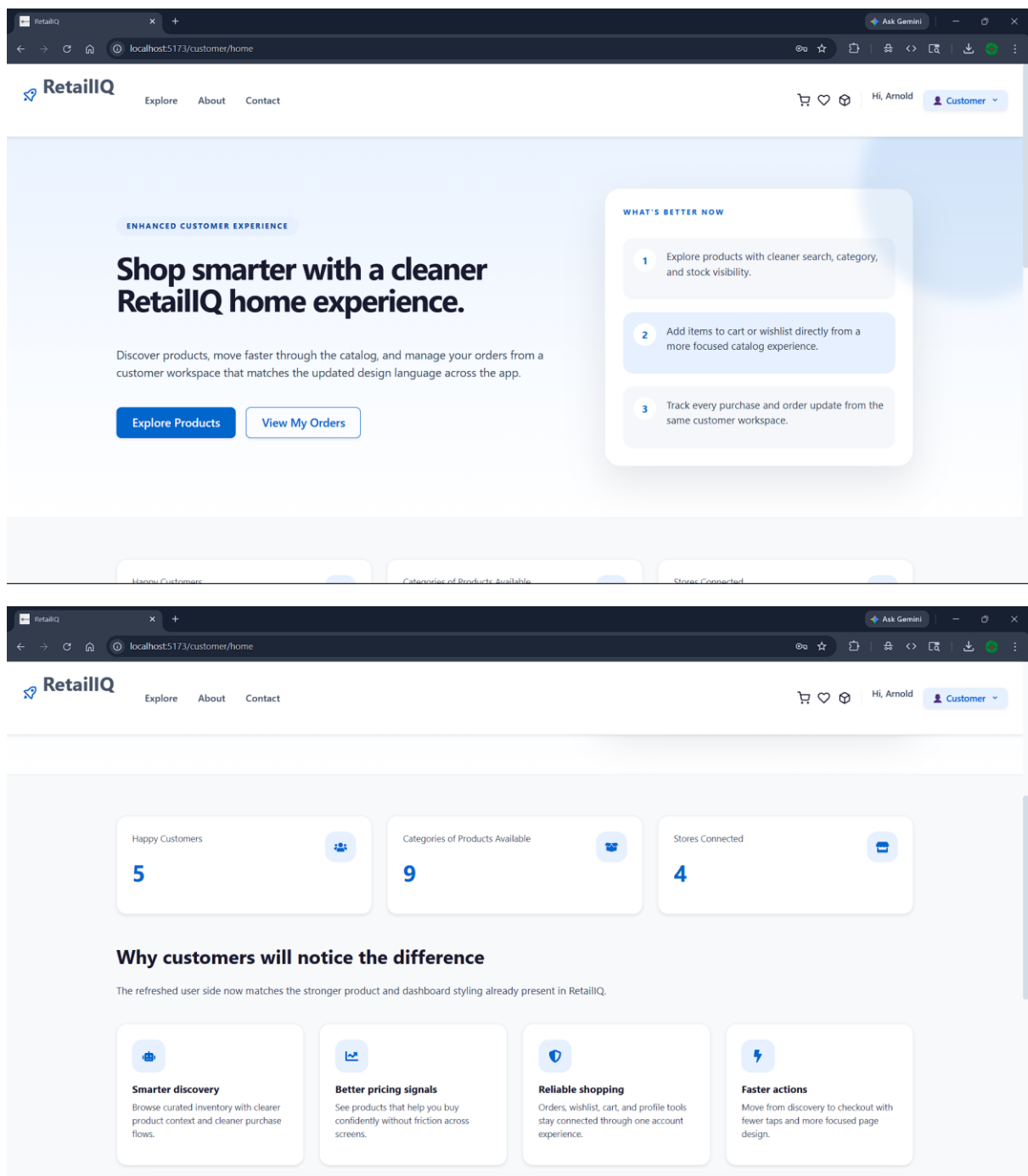
4.2. Screenshots

Login and signup screen

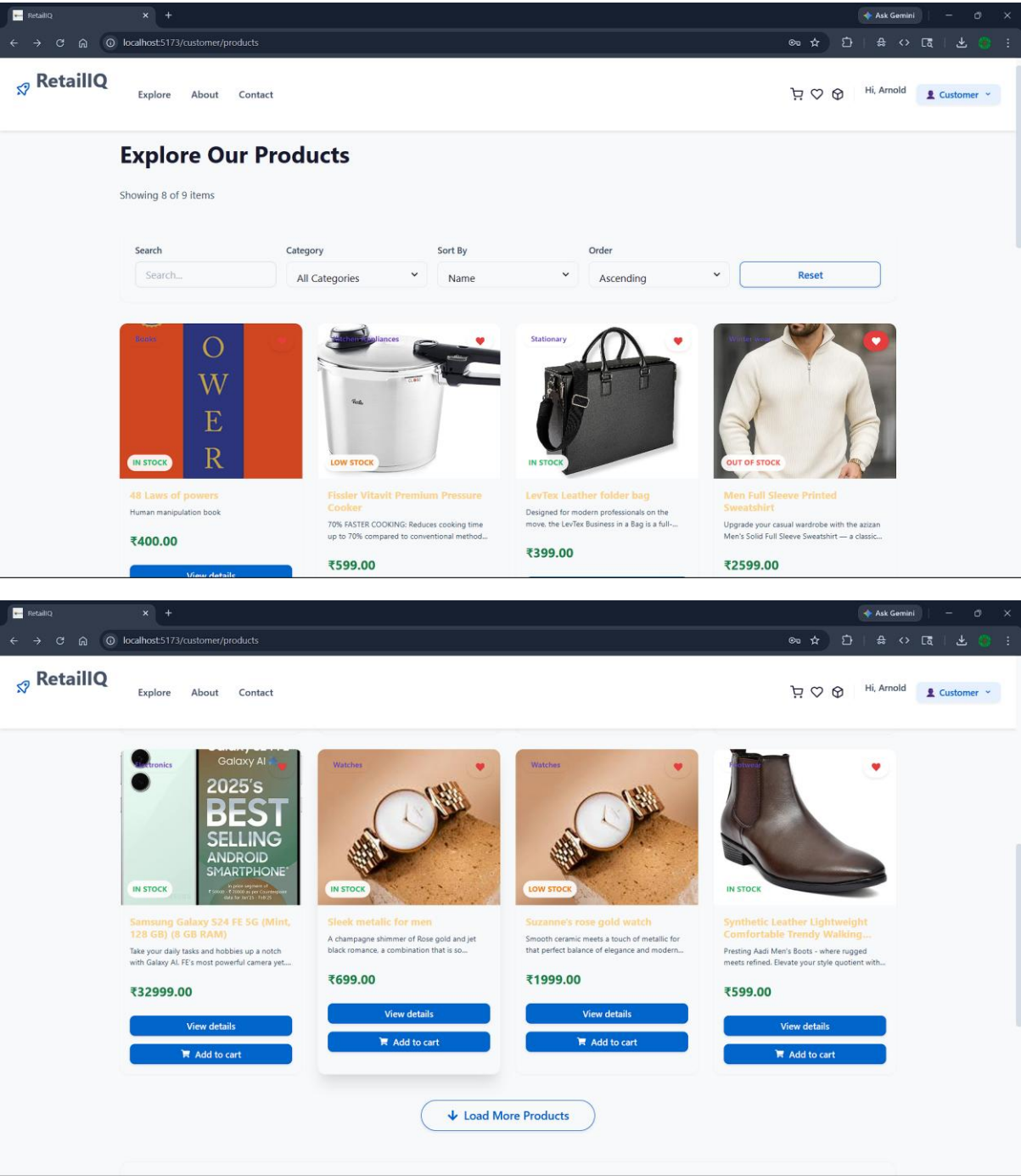
The first screenshot shows the 'ACCOUNT LOGIN' screen. On the left, a blue sidebar titled 'RETAILIQ CONTROL CENTER' contains the text 'Run your store with cleaner signals and faster decisions.' and a list of features: 'Unified access for admin, supplier, and customer roles', 'Real-time order visibility and product insights', and 'Faster workflows built around the updated RetailIQ design'. Below this is a 'RECOMMENDED' section advising users to sign in with their registered email or phone. The main white area is titled 'Sign in to RetailIQ' and includes a 'Forgot password?' link. It features input fields for 'Email or Phone' (with a placeholder 'you@example.com or +11234567890') and 'Password', followed by a 'Sign in' button and a link to 'Create an account' for new users. A footer note states, 'By signing in, you agree to our Terms & Privacy Policy'.

The second screenshot shows the 'NEW ACCOUNT' screen. The blue sidebar is titled 'RETAILIQ ONBOARDING' and says 'Create your account with the same clean workspace experience.' It lists features: 'Unified access across admin, supplier and other experiences', 'Access to product insights, orders, and profile management', and 'Streamlined onboarding aligned with the updated login design'. The main white area is titled 'Create your RetailIQ account' and asks users to 'Fill in your details to start using the updated RetailIQ platform.' It includes input fields for 'First name', 'Last name', 'Email' (placeholder 'you@example.com'), 'Phone' (placeholder '1234567890'), 'Gender' (a dropdown menu), and 'Date of Birth' (a date picker). A 'Password' field with a 'Minimum 8 characters' requirement and a toggle for visibility is also present. A 'Create account' button is at the bottom, with a link to 'Already have an account? Sign in' below it.

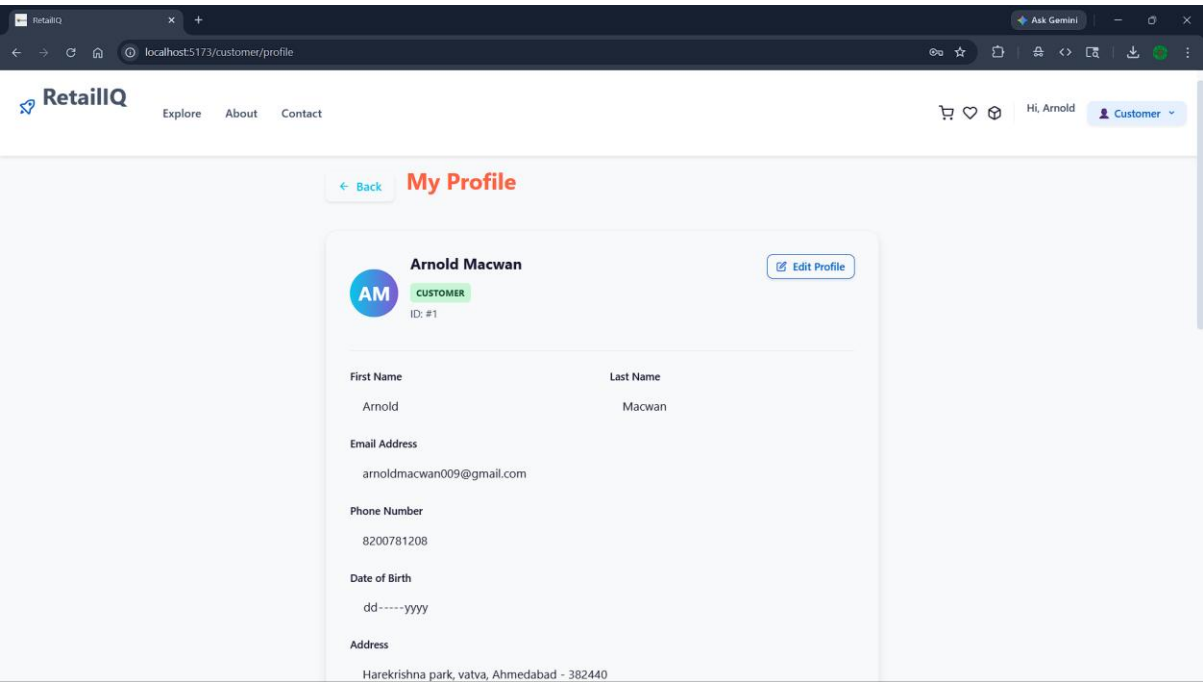
Customer Home page



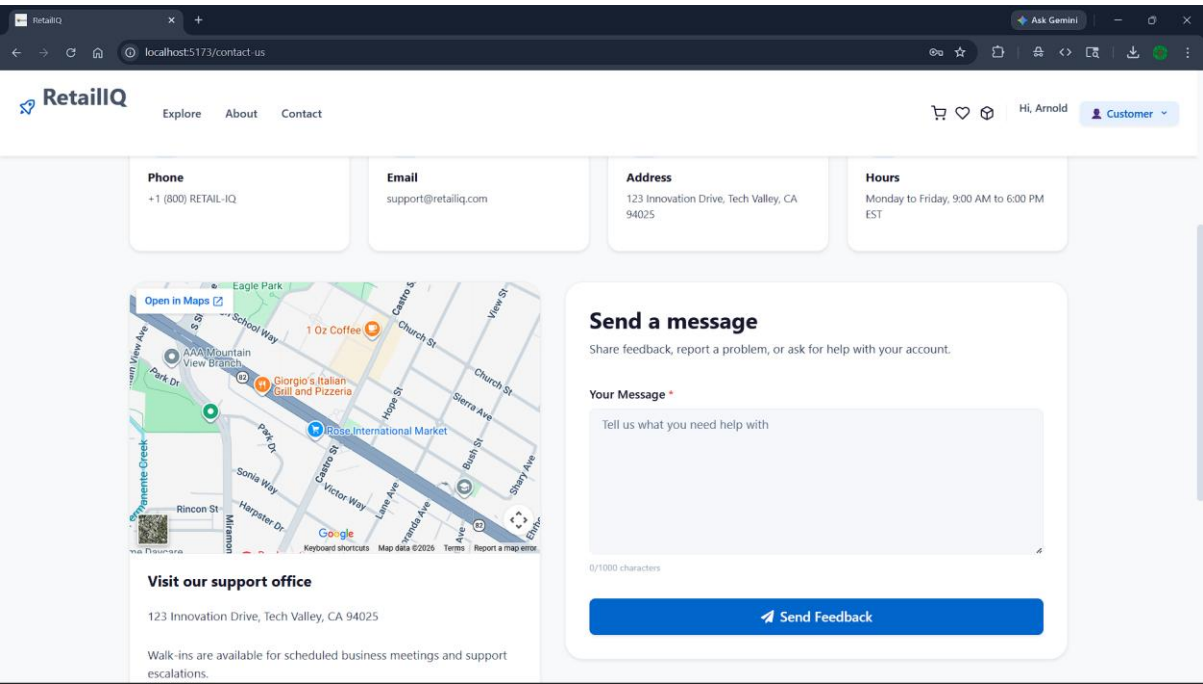
Product Catalog



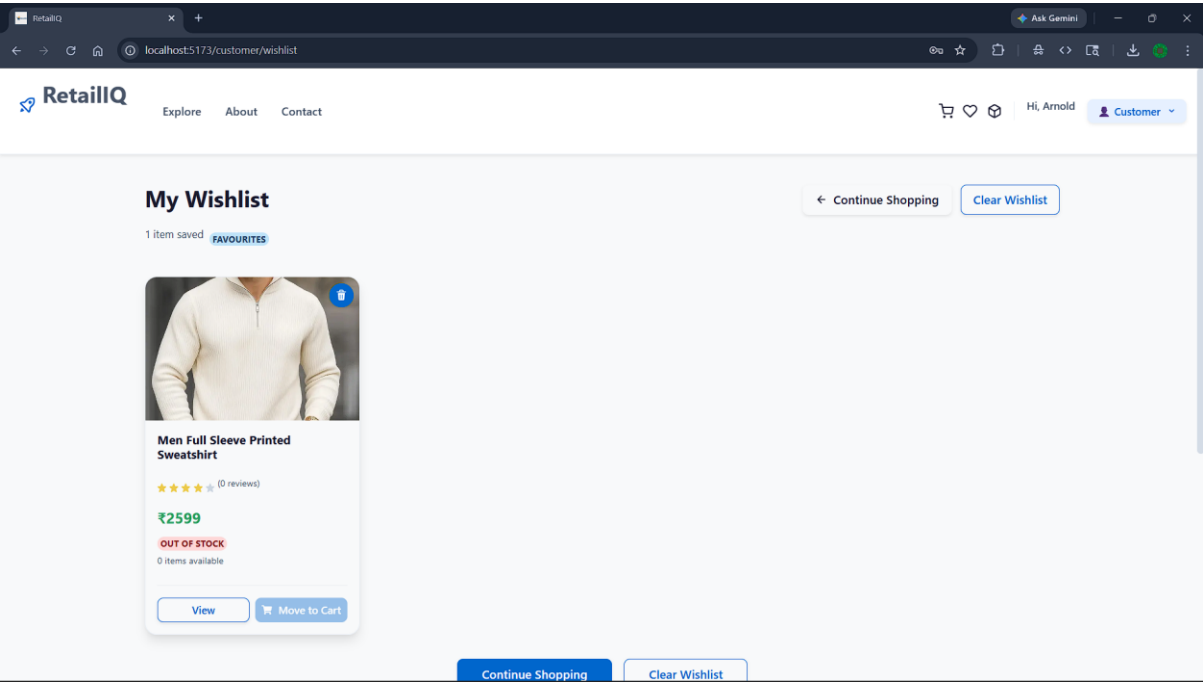
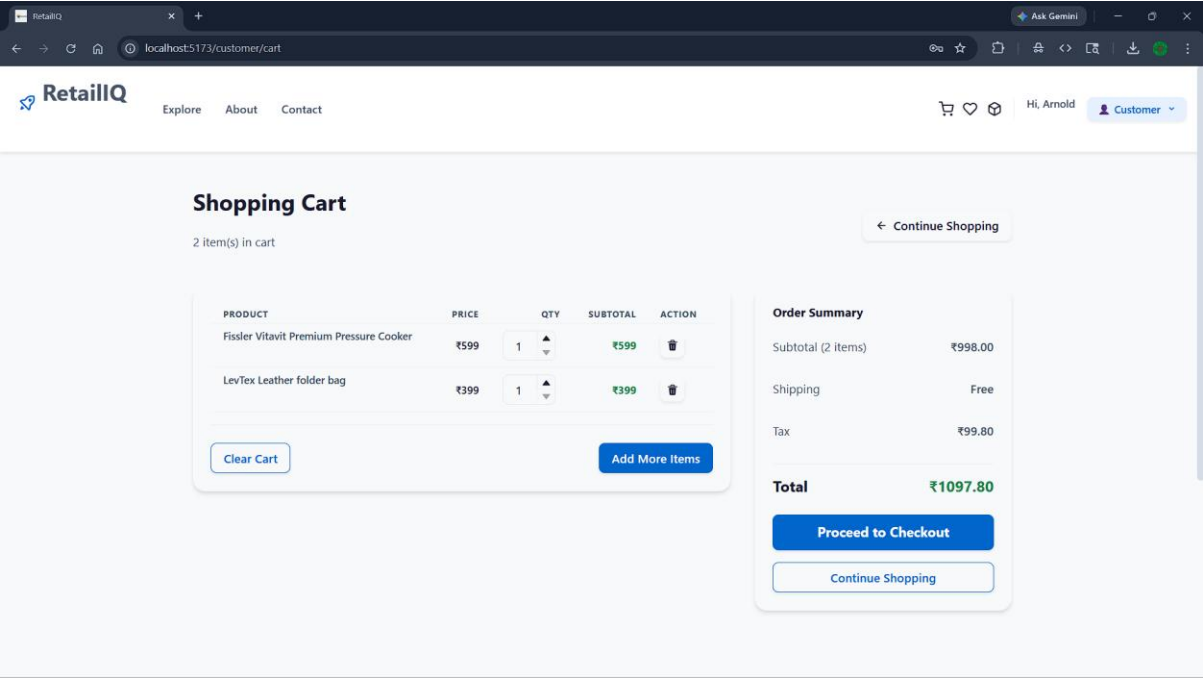
Profile Page



Feedback & contact us



Cart & Wishlist



Order history

RetailIQ

ExploreAboutContact

Hi, Arnold

Customer

My Orders

You have 8 orders

8

TOTAL ORDERS

2

COMPLETED

3

PROCESSING

2

CANCELLED

Back to Home

ORDER NO.	DATE	AMOUNT	STATUS	PAYMENT	ACTIONS
ORD-1773427327333-AVEFGH6V5	14/3/2026	₹1760.00	PROCESSING	PAID	View Cancel
ORD-1773427182917-OFQC55LLC	14/3/2026	₹1760.00	PENDING	PENDING	View Cancel
ORD-1773427102055-9E91G3SAZ	14/3/2026	₹1760.00	CANCELLED	PENDING	View
ORD-1767085270483-LEJFYYSZ	30/12/2025	₹31447.90	COMPLETED	PAID	View

Admin Dashboard

RetailIQ

Products

Hi, Virat

Admin

ADMIN PANEL

Dashboard

Users

Categories

Products

Stores

Store Managers

Feedbacks

Orders

Suppliers

Ask Suppliers

Admin Dashboard

Overview of system performance

Export

Analytics

Total Orders

9

Total Suppliers

3

Total Customers

9

Total Products

9

Recent Activity

ORD-1773427327333-AVEFGH6V5

customer_order • 3/14/2026, 12:12:07 AM

ORD-1773427182917-OFQC55LLC

customer_order • 3/14/2026,

Quick Actions

View Reports

Manage Users

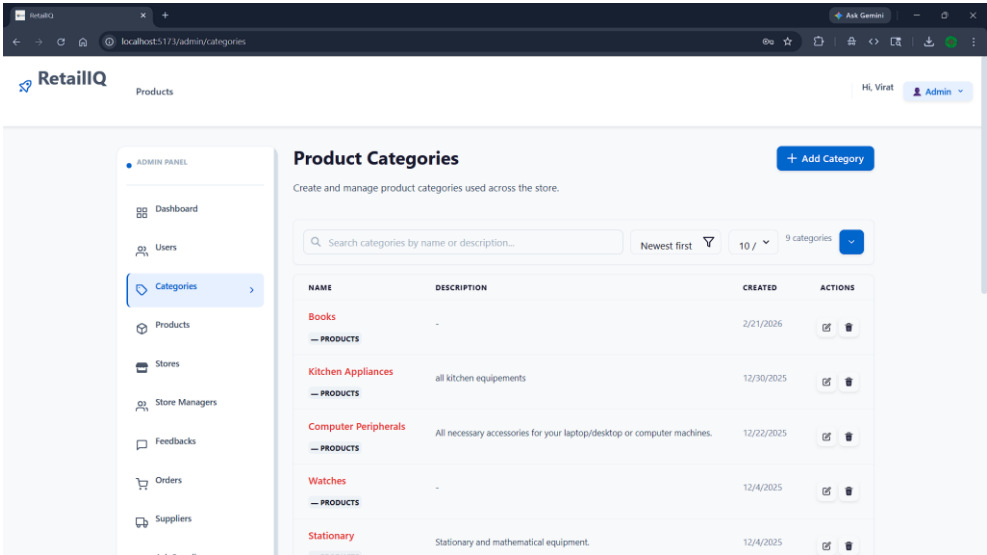
System Settings

Backup Data

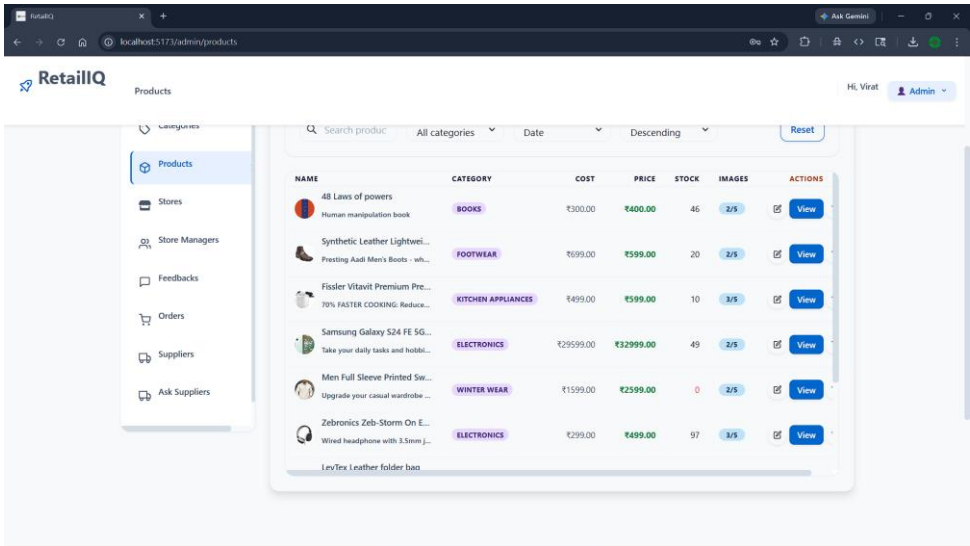
Summary

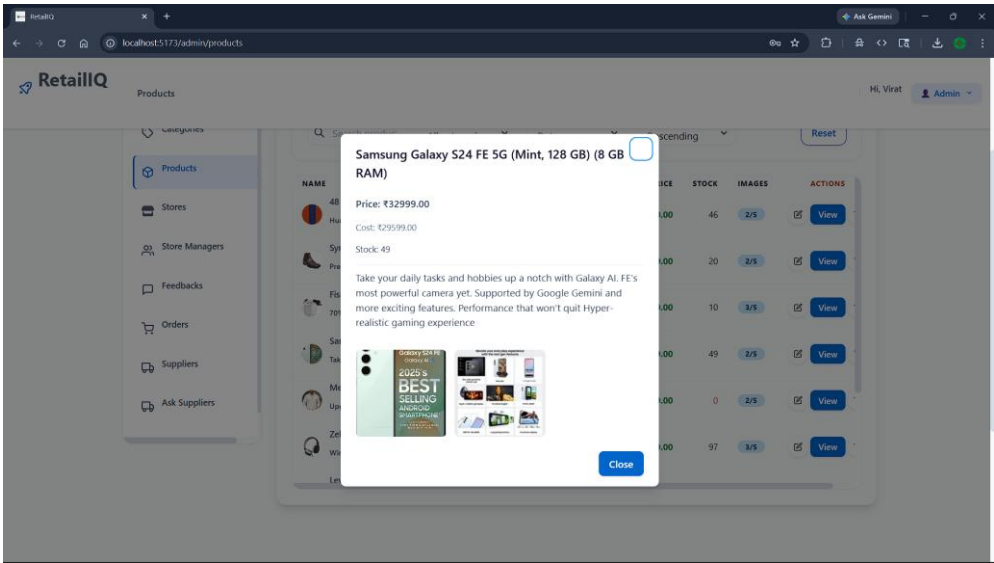
Quick summary or KPIs can go here.

Manage Product Categories by Admin

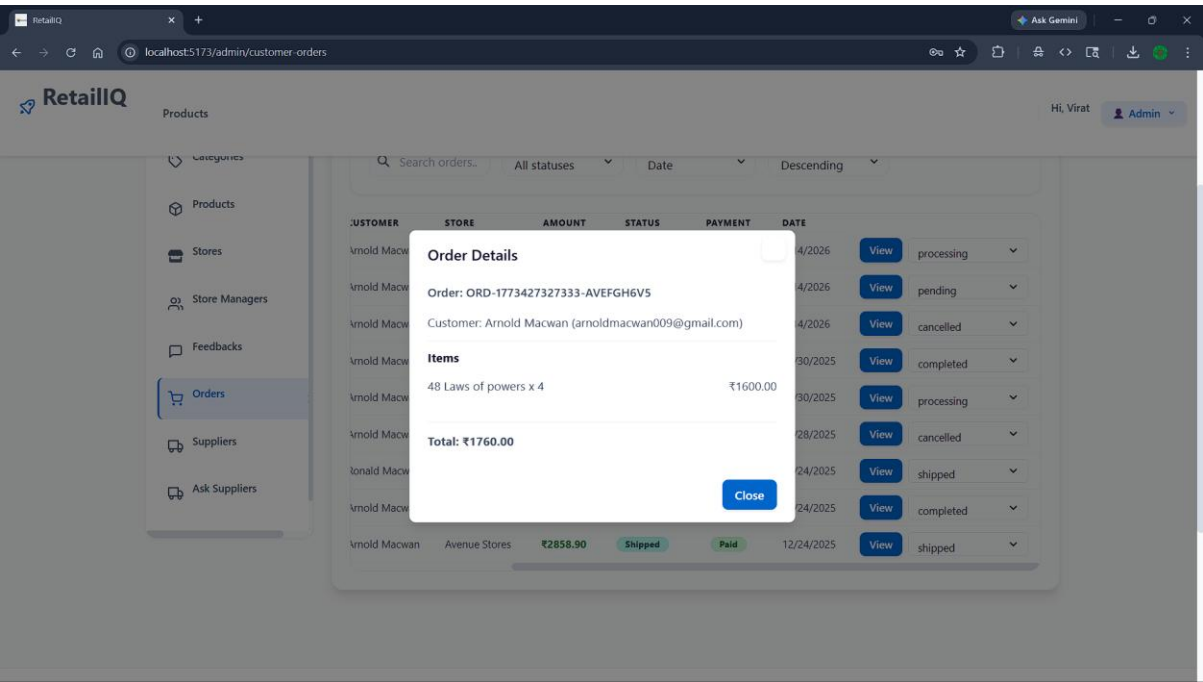


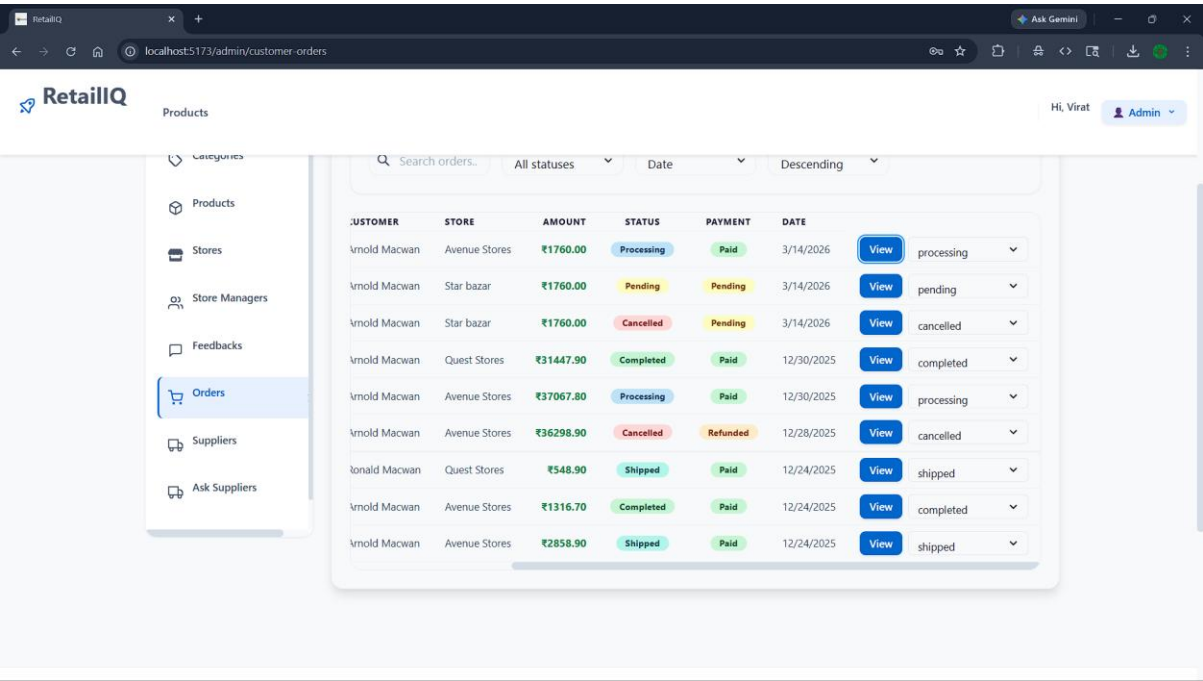
Manage and view products



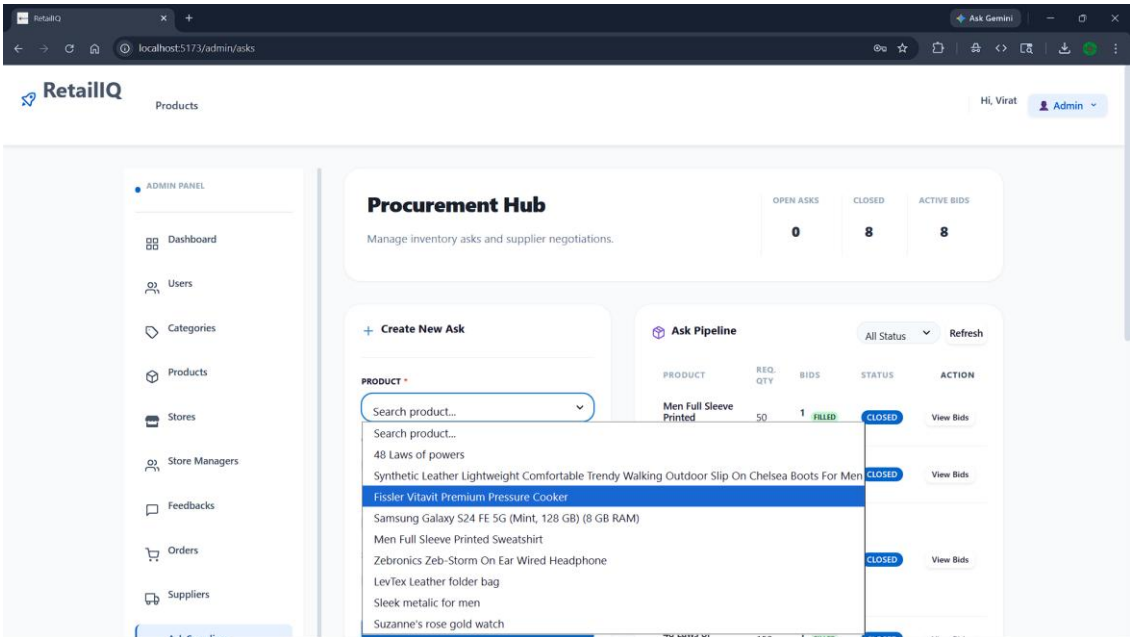


Manage Customer Orders

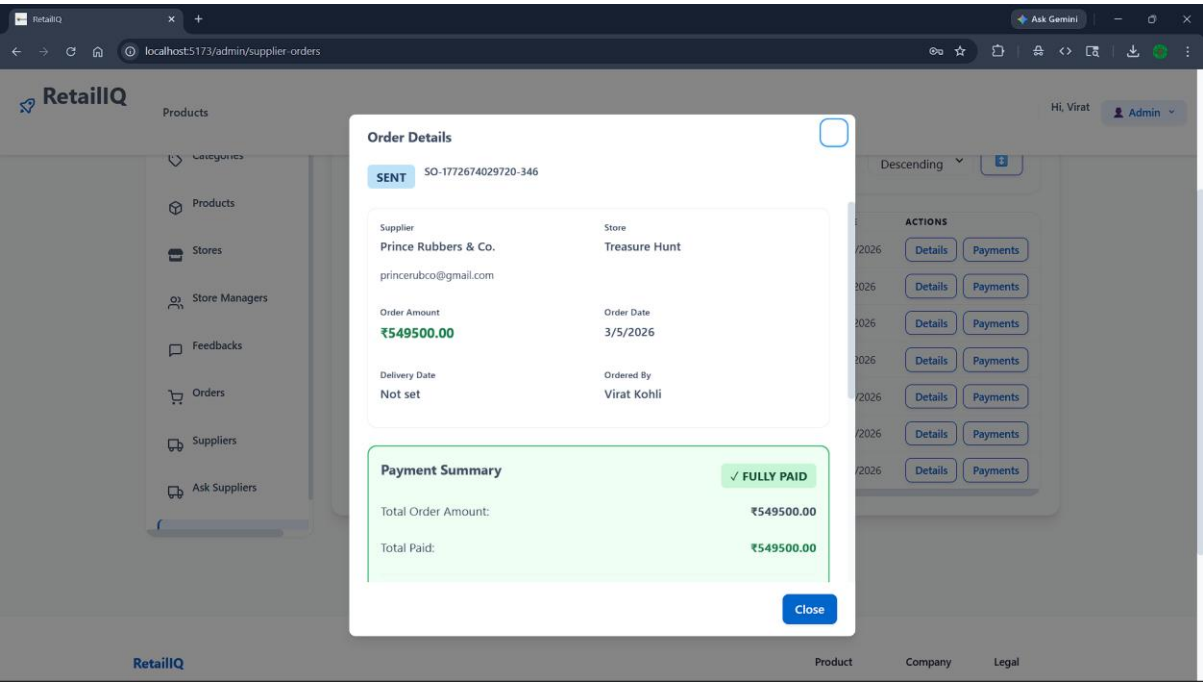




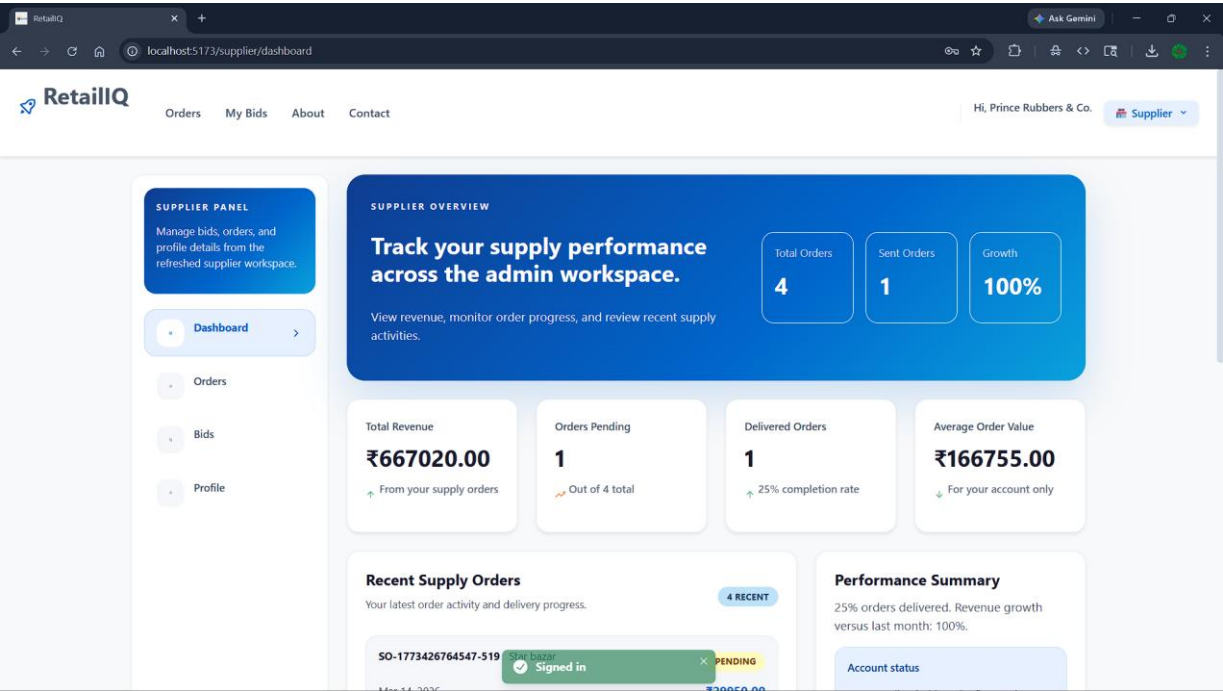
Ask for Bids

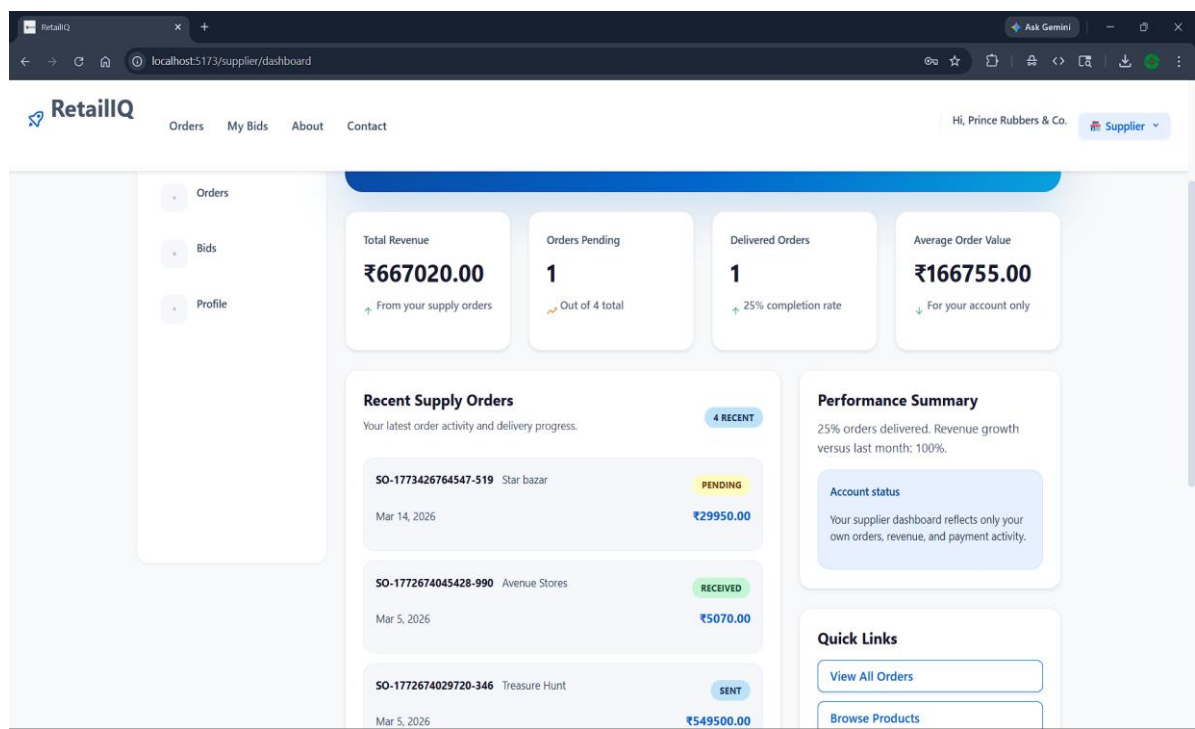


Manage Orders by admin

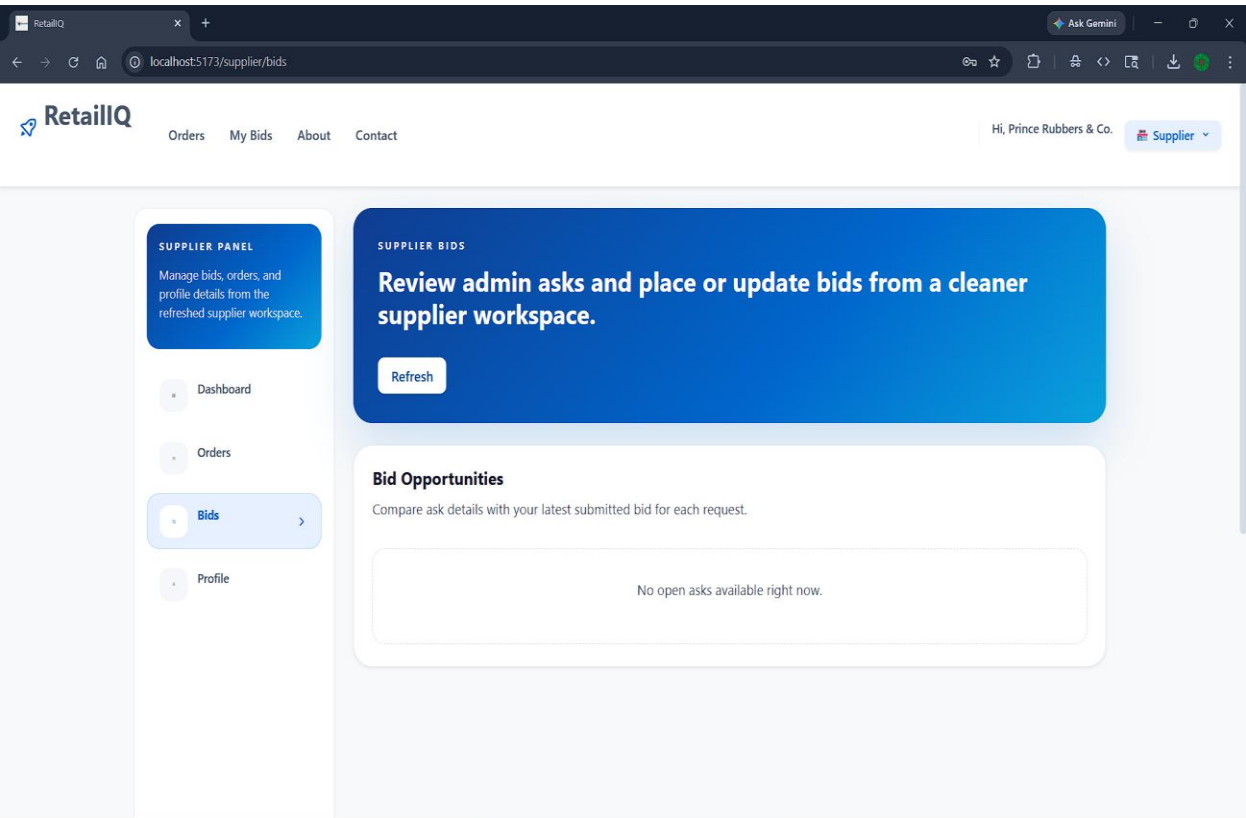


Supplier Dashboard





Supplier Biding



Supplier Order Summary

RetailIQ

Orders

My Bids

About

Contact

Hi, Prince Rubbers & Co.

Supplier

SUPPLIER PANEL

Manage bids, orders, and profile details from the refreshed supplier workspace.

Dashboard

Orders

Bids

Profile

ORDER DETAIL

SO-1772674045428-990

Supplier order detail view with the same cleaner structure used across the updated admin-style pages.

Order Summary

Store

Avenue Stores

Status

RECEIVED

Total

₹5070.00

Payment Overview

2 payment record(s) found for this order.

Items

PRODUCT	QTY	UNIT COST	TOTAL
Synthetic Leather Lightweight Comfortable Trendy Walking Outdoor Slip On Chelsea Boots For Men	30	₹169.00	₹5070.00

Payments

DATE	AMOUNT	METHOD	REF
3/5/2026	₹5070.00	CASH	-
3/5/2026	₹20.00	CASH	-

RetailIQ

Smart retail analytics & operations platform for local shops, supermarkets, and fashion stores.

Product

Company

Legal

Dashboard

About Us

Privacy Policy

Analytics

Blog

Terms of Service

Forecasting

Contact Us

Cookie Policy

Integration

Careers

Security

© 2026 RetailIQ. All rights reserved. | Made with for retail

40

5. Agile Documentation

5.1. Project Charter

Project Name	RetailIQ
Guide	Prof. Atik Sharma
Start Date	01-10-2025
End Date	30-04-2025
Project Scope	RetailIQ is a smart retail analytics and operations platform built to unify and automate key retail workflows. The project covers inventory management, sales tracking, supplier coordination, and AI-driven analytics for demand forecasting, market-basket insights, dynamic pricing, and predictive stock management.
Project Mission	To empower local retailers, supermarkets, and small businesses with intelligent, data-driven tools that simplify operations, optimize inventory, enhance profitability, and enable smarter decision-making through actionable insights and modern AI/ML technologies.
Project Vision	To become the leading smart retail analytics platform that transforms traditional retail into efficient, tech-enabled businesses—bridging the gap between small stores and enterprise-level retail intelligence through affordable, scalable, and automated solutions.

5.2. Project Roadmap

	October		November		December		January		February		February	
Tasks	Week 01 - 02	Week 03 - 04	Week 01 - 02	Week 03 - 04	Week 01-02	Week 01-02	Week 03-04	Week 01-02	Week 01 - 02	Week 03 - 04	Week 01-02	Week 03-04
Project Setup & Architecture Design	✓											
Database Finalization & ER Diagram	✓											
Authentication & RBAC		✓										
Store & User Management Module			✓		✓							
Product & Category Management			✓									
Supplier & Purchase Orders Module				✓		✓						
Inventory Management					✓							
Customer Orders & Billing					✓	✓		✓				
Invoice Module Integration								✓				
Payment Integration				✓			✓					
Analytics Dashboard (Basic Reports)							✓		✓			
Demand Forecasting Model											✓	✓
Market Basket Analysis										✓		
Dynamic Pricing Engine								✓	✓			
Stock-out & Overstock Prediction											✓	
UI Enhancements & Performance Optimization		✓				✓			✓			

5.3. User Story

Sr.	As a	I want to	So that
1	Admin	create and manage user accounts	I can control system access
2	Admin	assign roles (admin, store manager, supplier, customer)	users have proper access permissions
3	Admin	add and update store details	store information remains accurate
4	Admin	add new product categories	products can be organized properly

5	Admin	add and update products	inventory remains up to date
6	Admin	upload product images	customers can visually identify products
7	Admin	set reorder levels for products	I can prevent stock-outs
8	Admin	track available stock in real-time	I can monitor inventory status
9	Admin	receive low-stock alerts	I can reorder products on time
10	Admin	create supply orders	I can restock inventory efficiently
11	Admin	track supply order status	I know when products will arrive
12	Admin	record supplier payments	financial records remain accurate
13	Supplier	view supply orders assigned to me	I can fulfil orders promptly
18	Customer	view available products	I can decide what to purchase
19	Customer	add products to cart	I can manage my purchases before checkout
20	Customer	place an order	I can complete my purchase
21	Customer	make payments via multiple methods	I can choose convenient payment options
22	Customer	view my order history	I can track my past purchases
23	Customer	create Wishlist	I can save products for later
24	Admin	view sales dashboard	I can monitor overall business performance
32	Admin	export sales and inventory reports	I can share data for accounting or auditing
33	Admin	monitor supplier performance ratings	I can choose reliable vendors
34	User	reset password securely	I can regain access if I forget credentials
35	Admin	view system activity logs	I can monitor system usage and security

5.4. Release Plan

Task	Start Date	End Date	Status	Release Date
Database Setup & Initial Configuration	01-10-2025	03-10-2025	Released	04-10-2025
Authentication System (Login / Signup)	04-10-2025	08-10-2025	Released	10-10-2025
Password Reset & Email Verification	09-10-2025	12-10-2025	Released	14-10-2025
Role-Based Access Control (RBAC)	13-10-2025	17-10-2025	Released	19-10-2025
User Profile Management	18-10-2025	22-10-2025	Released	24-10-2025
Store Management Module	23-10-2025	28-10-2025	Released	30-10-2025
Product Categories Management	01-11-2025	05-11-2025	Released	07-11-2025
Product Management Module	06-11-2025	12-11-2025	Released	14-11-2025
Supplier Management Module	13-11-2025	18-11-2025	Released	20-11-2025
Inventory Management System	19-11-2025	26-11-2025	Released	28-11-2025
Supply Orders (Purchase Orders)	27-11-2025	03-12-2025	Released	05-12-2025
Supply Payments Tracking	04-12-2025	08-12-2025	Released	10-12-2025
Customer Orders (Sales Module)	09-12-2025	16-12-2025	Released	18-12-2025
Invoice Generation Module	17-12-2025	22-12-2025	Released	24-12-2025
Customer Payment Integration	26-12-2025	31-12-2025	Released	02-01-2026

Wishlist & Cart System	03-01-2026	08-01-2026	Released	10-01-2026
Feedback & Ratings System	09-01-2026	13-01-2026	Released	15-01-2026
Sales Analytics Dashboard	18-01-2026	24-01-2026	Released	27-01-2026
Demand Forecasting (ML Model)	01-02-2026	08-02-2026	Not Released	-
Market Basket Analysis	09-02-2026	15-02-2026	Not Released	-
Dynamic Pricing Recommendation	16-02-2026	22-02-2026	Not Released	-
Stock-out & Overstock Prediction	23-02-2026	01-03-2026	Not Released	-
Final System Testing & Bug Fixing	02-03-2026	10-03-2026	Not Released	-
Deployment & Documentation	11-03-2026	20-03-2026	Not Released	-
Final Project Review & Release	21-03-2026	31-03-2026	Not Released	-

5.5. Sprint/Backlog plan

Id	Task/Module	Assigned to	Status	Estimated Time (hours)	Story Points
1	Database Setup & Initial Configuration	Backend Dev	Done	6	3
2	Authentication System (Login / Signup)	Backend Dev	Done	10	5
3	Password Reset & Email Verification	Backend Dev	Done	8	3
4	Role-Based Access Control (RBAC)	Backend Dev	Done	10	5
5	User Profile Management	Full Stack Dev	Done	8	3

6	Store Management Module	Full Stack Dev	Done	12	5
7	Product Categories Management	Backend Dev	Done	6	2
8	Product Management Module	Full Stack Dev	Done	14	8
9	Supplier Management Module	Backend Dev	Done	10	5
10	Inventory Management System	Full Stack Dev	Done	16	8
11	Supply Orders (Purchase Orders)	Backend Dev	Done	12	5
12	Supply Payments Tracking	Backend Dev	Done	8	3
13	Customer Orders (Sales Module)	Full Stack Dev	Done	16	8
14	Invoice Generation Module	Backend Dev	Done	10	5
15	Customer Payment Integration	Backend Dev	Done	12	5
16	Wishlist & Cart System	Frontend Dev	Done	10	5
17	Feedback & Ratings System	Frontend Dev	Done	6	2
18	Sales Analytics Dashboard	Data/Frontend Dev	Done	14	8
19	Demand Forecasting (ML Model)	Data Scientist	In Progress	18	13
20	Market Basket Analysis	Data Scientist	In Progress	16	8
21	Dynamic Pricing Recommendation	Data Scientist	Pending	16	8
22	Stock-out & Overstock Prediction	Data Scientist	Pending	18	13
23	API Optimization & Performance	Backend Dev	Pending	10	5

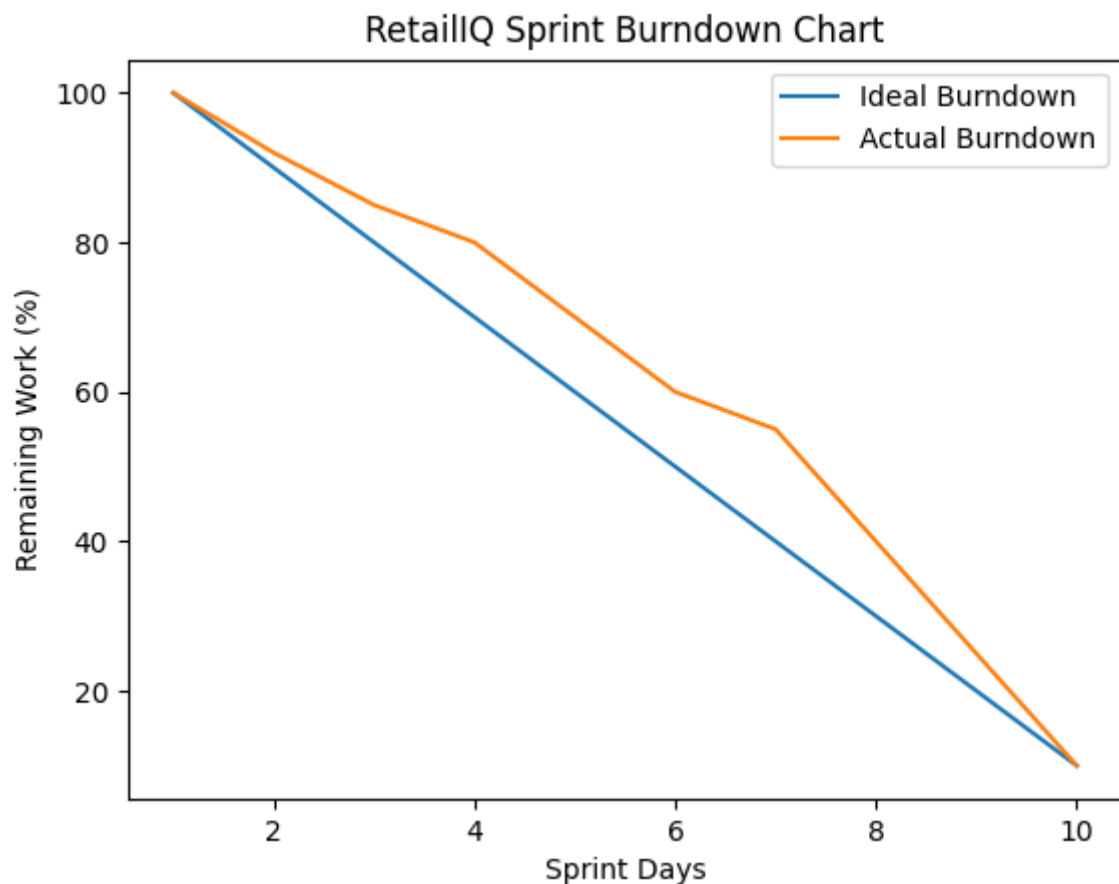
24	UI Enhancements & UX Improvements	Frontend Dev	Pending	12	5
25	System Integration Testing	QA Tester	Pending	14	8
26	Bug Fixing & Stabilization	Full Team	Pending	16	8
27	Deployment Setup (Server/Cloud)	DevOps	Pending	10	5
28	Documentation (SRS, Reports)	Full Stack Dev	Pending	12	5
29	Final System Review	Team Lead	Pending	8	3
30	Final Project Submission & Demo Prep	Full Team	Pending	10	5

5.6. Test Plan

Test#	Action	Expected Result	Actual Result	Status
1	Register a new user	User account is created successfully	True	Pass
2	Login with valid credentials	User is logged in and redirected to dashboard	True	Pass
3	Login with invalid credentials	Error message is displayed	True	Pass
4	Reset password via email	Password reset link is sent	True	Pass
5	Verify OTP for authentication	OTP is validated successfully	True	Pass
6	Update user profile details	Profile is updated successfully	True	Pass
7	Add a new store	Store is created and saved in database	True	Pass
8	Add product category	Category is added successfully	True	Pass
9	Add new product	Product is stored with correct details	True	Pass
10	Update product details	Product information is updated	True	Pass
11	Delete product	Product is removed from system	True	Pass
12	Upload product image	Image is uploaded and linked to product	True	Pass
13	Add supplier	Supplier details are stored successfully	True	Pass
14	Create supply order	Supply order is generated with correct details	True	Pass

15	Update supply order status	Status changes successfully	True	Pass
16	Record supplier payment	Payment is stored and linked to supplier	True	Pass
17	Check inventory stock update	Stock levels update after supply/order	True	Pass
18	Receive low-stock alert	System shows alert for low inventory	True	Pass
19	Create customer order	Order is successfully created	True	Pass
20	Add items to order	Items are added with correct pricing	True	Pass
21	Apply discount on order	Discount is applied correctly	True	Pass
22	Generate invoice	Invoice is generated with correct details	True	Pass
23	Process customer payment	Payment status updates to “paid”	True	Pass
24	Add product to cart	Product is added to cart successfully	True	Pass
25	Remove product from cart	Product is removed from cart	True	Pass
26	Add product to wishlist	Product is saved in wishlist	True	Pass
27	Submit feedback	Feedback is recorded successfully	True	Pass
28	View sales dashboard	Dashboard displays correct analytics	True	Pass
29	Generate sales report	Report shows accurate data	True	Pass
30	Run demand forecasting model	Predictions are generated successfully	True	Pass
31	Perform market basket analysis	Product associations are displayed	True	Pass
32	Get dynamic pricing suggestion	Pricing recommendations are shown	False	Fail
33	Detect stock-out prediction	System predicts low-stock scenarios	False	Fail
34	Export reports (CSV/Excel)	File downloads successfully	True	Pass
35	Logout from system	User is logged out securely	True	Pass

5.7. Sprint burn chart (estimated)



6. Proposed Enhancement

While inventory management, supplier tracking, sales monitoring, demand forecasting, and market basket analysis are just a few of the features that RetailIQ offers as an intelligent retail operations and analytics platform, there are a number of improvements that could be added in later iterations to increase the system's functionality and boost its effectiveness, scalability, and usability.

- Multi-store management
- Barcode & QR code integration
- POS integration
- Customer Rewards program
- Automated Order Generation

7. Conclusion

The development of RetailIQ — Smart Retail Operations & Analytics Platform demonstrates how modern information technology and data analytics can significantly improve the efficiency and decision-making capabilities of retail businesses. The system integrates core retail management functionalities such as inventory tracking, supplier management, sales monitoring, customer orders, and invoicing within a unified platform. By centralizing these operations, RetailIQ simplifies daily retail activities and reduces the dependency on manual record keeping.

8. Project Learnings

The development of the RetailIQ project provided valuable insights into how data and technology can transform the retail industry. It helped us understand the practical implementation of analytics, customer behavior tracking, and intelligent decision-making systems. Throughout the project, we gained both technical knowledge and problem-solving skills by working on real-world retail scenarios.

- Data-driven decision making
- Developed skills on data visualization on live data
- Role of AI and analytics in modern retail systems
- Improved team collaboration and management
- Faced challenges like data inconsistency and live debugging

9. Bibliography

- Reports & Industry Sources
 - McKinsey & Company. (2023). The future of retail: AI, personalization, and analytics.
 - Deloitte. (2024). Global Powers of Retailing Report.
 - PwC. (2023). Retail and Consumer Insights Report.

<https://retailiq.kantar.com/iq>

<https://www.bluecore.com/retail-intelligence-solution/>